The empirical recurrence for OEIS A232427: recovering a conjecture that is not written down

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7 September 2026

Abstract

OEIS A232427 records "Empirical recurrence of order 55", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 90$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 55, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 55 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A232427 is "Number of (n+1)X(3+1) 0..2 arrays with every element next to itself plus and minus one within the range 0..2 horizontally or vertically.". It has offset 1 and begins

$$64, 1632, 33708, 650696, 14465384, 313201804, 6514702216, 139122934752, \ \ldots$$

The entry states:

> Empirical recurrence of order 55 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 07:26:21, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 6642$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 90$ of the 6642, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 90$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 13284$ exact terms of the model returns order 55 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{55} c_i \, a(n-i),$$

c_1	2	c_{15}	1780127760828	c_{29}	−30421158231367807 56	c_{43}	−2349282416049271603
c_2	195	c_{16}	250826575256	c_{30}	−1305147459968205275 2	c_{44}	44703901390340725145
c_3	3643	c_{17}	−8156206468508	c_{31}	5268352163232943174 4	c_{45}	−5209875161480817541
c_4	28296	c_{18}	−27491849271263	c_{32}	−1258263499696914367 04	c_{46}	18136074777525433139
c_5	−78485	c_{19}	−854187834337302	c_{33}	870657901168982830 72	c_{47}	−1068240994510191984
c_6	−1851855	c_{20}	582032295100542	c_{34}	151549966916888108864	c_{48}	−1695827978717075865
c_7	−5597960	c_{21}	−852351970473719	c_{35}	−511234775769211060224	c_{49}	−909464728641103462
c_8	−14545543	c_{22}	−27911638173283669	c_{36}	779878366446923160320	c_{50}	−3009410383032090624
c_9	−102658236	c_{23}	81183204558933944	c_{37}	−521977610515420060160	c_{51}	−165828488012077465
c_{10}	828734107	c_{24}	−302990433052227	c_{38}	−114287652274122905600	c_{52}	5177768480476233728
c_{11}	3664842685	c_{25}	−223865081884433873	c_{39}	966583193381665372160	c_{53}	686517468197289984
c_{12}	−680110694	c_{26}	1451581827739577131	c_{40}	−1362548282115281539072	c_{54}	145522562959409152
c_{13}	108343322251	c_{27}	−1310363694069039698	c_{41}	1110790873707815043072	c_{55}	18014398509481984
c_{14}	539232374557	c_{28}	712629056767260864	c_{42}	−436331951401826959360		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 55, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $55 + 4$ terms removed returns order 55 as well, so $\deg q_{\min} = 55 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ☐

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $55 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 55 that this sequence satisfies — and that family turns out to have exactly one member.

References

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